

QTM 220 Final Practice Problems

Problem 1

Suppose you have a sample $Y_1, Y_2, \dots, Y_n$ drawn with replacement from a population with mean μ and standard deviation σ .

Consider the following estimator for μ : $\tilde{Y} = \frac{1}{n+2} \sum_{i=1}^n Y_i$

Part A

What is $E[\tilde{Y}]$?

$$\begin{aligned} E[\tilde{Y}] &= E\left[\frac{1}{n+2} \sum_{i=1}^n Y_i\right] \\ &= \frac{1}{n+2} E\left[\sum_{i=1}^n Y_i\right] \\ &= \frac{1}{n+2} \sum_{i=1}^n E[Y_i] \end{aligned}$$

by linearity of expectations

$$= \frac{1}{n+2} \sum_{i=1}^n E[Y]$$

by IID

$$= \frac{1}{n+2} \sum_{i=1}^n \mu$$

by definition

$$= \frac{1}{n+2} n\mu$$

$$= \frac{n}{n+2} \mu$$

Part B

How does $E[\tilde{Y}]$ compare to $E[\bar{Y}]$? (where $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$)

similar steps show that $E[\bar{Y}] = \mu$. $\bar{Y}$ is unbiased, $\tilde{Y}$ is biased, but $\lim_{n \rightarrow \infty} \frac{n}{n+2} = 1$ so bias will decrease as n increases.

Part C

Showing your work, calculate $Var[\tilde{Y}]$.

$$Var[\tilde{Y}] = V[\frac{1}{n+2} \sum_{i=1}^n Y_i]$$

$$= (\frac{1}{n+2})^2 V[\sum_{i=1}^n Y_i]$$

$$= (\frac{1}{n+2})^2 \sum_{i=1}^n V[Y]$$

because IID, covariances are 0

$$= (\frac{1}{n+2})^2 \sum_{i=1}^n \sigma^2$$

by definition

$$= (\frac{1}{n+2})^2 n\sigma^2$$

$$= \frac{n}{(n+2)^2} \sigma^2$$

Part D

How does $Var[\tilde{Y}]$ compare to $Var[\bar{Y}]$?

Similar steps show that $V[\tilde{Y}] = \frac{\sigma^2}{n}$

$\frac{1}{n} > \frac{n}{(n+2)^2}$ so our $\tilde{Y}$ estimator has a lower variance than the mean. But the estimator is biased, so we probably wouldn't want to use it over the mean.

Problem 2

Suppose we have a sample of $Y_1 \dots Y_n$ and $X_1 \dots X_n$ stored in an **R** dataframe 'df' with n rows. We run this code.

```
mod<- lm(y~x, data=df)
summary(mod)
```

Call:

```
lm(formula = y ~ x, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.2190	-0.8373	-0.0159	0.8350	4.1442

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.58927	0.14861	-3.965	0.000102 ***
x	0.33179	0.04095	8.102	5.48e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.31 on 198 degrees of freedom

Multiple R-squared: 0.249, Adjusted R-squared: 0.2452

F-statistic: 65.64 on 1 and 198 DF, p-value: 5.481e-14

```
B=10000
beta_0<- rep(NA,B)
beta_1<- rep(NA,B)
for( b in 1:B ){
```

```

bootind = sample(1:n,size=n,replace=T)
dfboot = df[bootind,]
mod<- lm(y~x, data=dfboot)
beta_0[b] = mod$coef[1]
beta_1[b] = mod$coef[2]
}

```

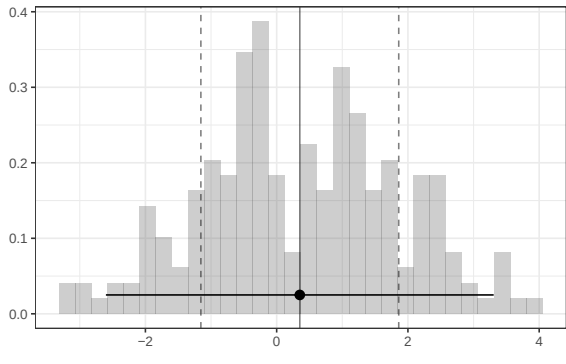


Figure 1: Histogram of the vector Y's elements

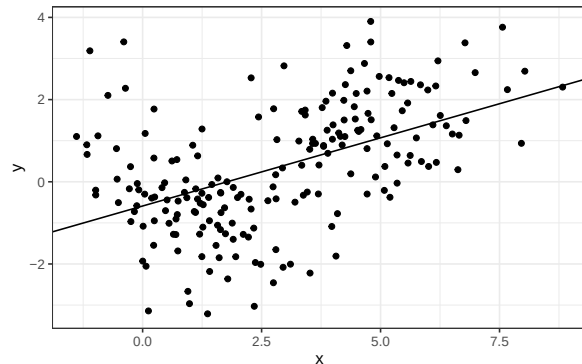


Figure 2: Scatterplot of X and Y

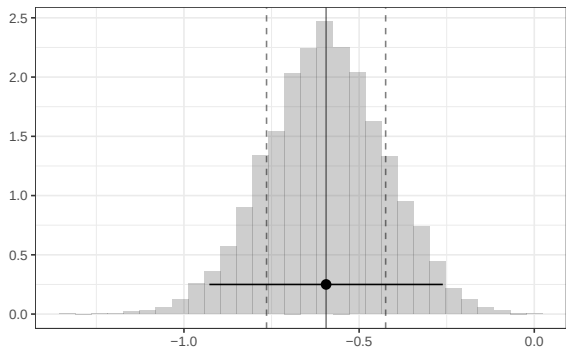


Figure 3: Histogram of the beta_0's elements

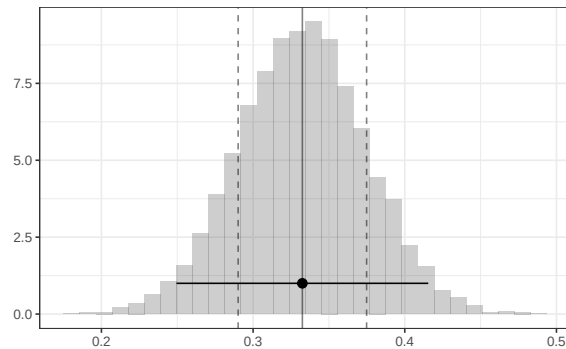


Figure 4: Histogram of the beta_1's elements

Part a

What does this code calculate? State it in words and mark it with an 'a', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y)
```

The sample mean. It's indicated by the solid vertical line on the sample histogram.

Part b

What does this code calculate? State it in words and mark it with a ‘b’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(Y)
```

The sample standard deviation of Y. It's the distance between the solid vertical and either dashed vertical line on the sample histogram.

Part c

What does this code calculate? State it in words and mark it with a ‘c’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mod<- lm(y~x, data=df)  
mod$coef[1]
```

The intercept of the simple regression of y on x. It is the intercept of the line on the scatterplot

Part d

What does this code calculate? State it in words and mark it with a ‘d’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(beta_1)
```

The bootstrap sampling distribution's standard deviation for the coefficient on x from the simple regression (slope). Estimated standard error. It's the distance between the solid vertical and either dashed vertical line on the beta_1 sampling distribution histogram.

Part e

What does this code calculate? State it in words and mark it with a ‘e’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mod$coef[2] + sqrt(diag(solve(t(X)%*%X)%*%t(X)%*%diag(resid^2)%*%X)%*%solve(t(X)%*%X)))[2]  
mod$coef[2] - sqrt(diag(solve(t(X)%*%X)%*%t(X)%*%diag(resid^2)%*%X)%*%solve(t(X)%*%X)))[2]
```

A 95% confidence interval — based on normal approximation — for the least squares slope from the simple regression of y on x for the population from which this sample was drawn. This is indicated by the horizontal segment (with a dot in the middle) on the sampling distribution histogram (`beta_1`).

Part f

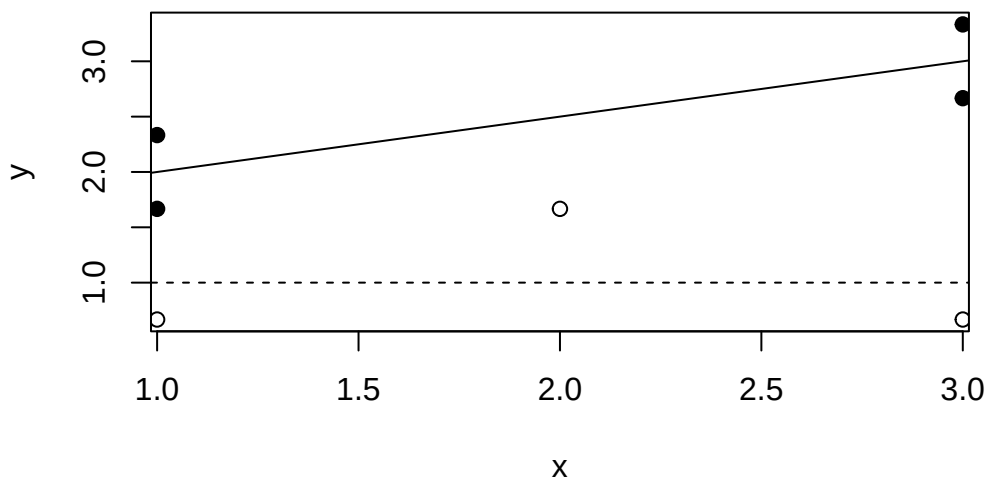
How could you calculate roughly the same thing as in part e using `sd(beta_1)`

We can use the standard deviation of the bootstrap sampling distribution as our estimate of standard error.

```
mod$coef[2] - 1.96 * sd(beta_1)
mod$coef[2] + 1.96 * sd(beta_1)
```

Problem 3

Consider the following data set with treatment d , covariate x , and outcome y , which has been fit with a linear interactive model. Note that the treated units with $d = 1$ are denoted as solid points, while the control units with $d = 0$ are denoted with open circles. The regression line for the treated units is solid, while the regression line for the control units is dashed.



Part A

What is the $\widehat{ATE}$ based on this model and data? Show your calculations.

$$\frac{1}{7}[(\frac{5}{3} - 1) + (\frac{8}{3} - 1) + (\frac{8}{3} - 1) + (\frac{10}{3} - 1) + (2 - \frac{2}{3}) + (2.5 - \frac{5}{3}) + (3 - \frac{2}{3})] = 1.547619$$

Part B

What is the $\widehat{ATT}$ based on this model and data? Show your calculations.

$$Y_i(1) - \hat{Y}_i(0)|d_i = 1$$

$$\frac{1}{4}[(\frac{5}{3} - 1) + (\frac{8}{3} - 1) + (\frac{8}{3} - 1) + (\frac{10}{3} - 1)] = \frac{6.333}{4} = 1.58333$$

Part B

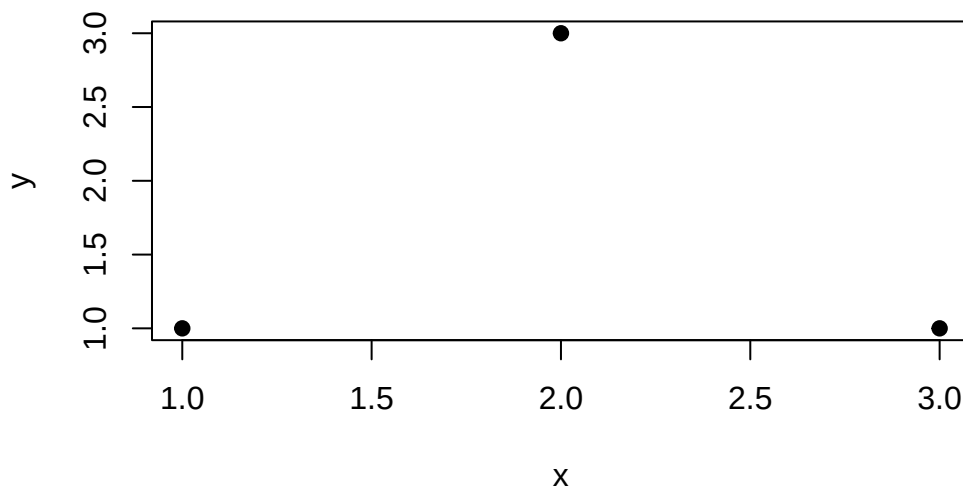
What is the $\widehat{ATC}$ based on this model and data? Show your calculations.

$$\hat{Y}_i(1) - Y_i(0)|d_i = 0$$

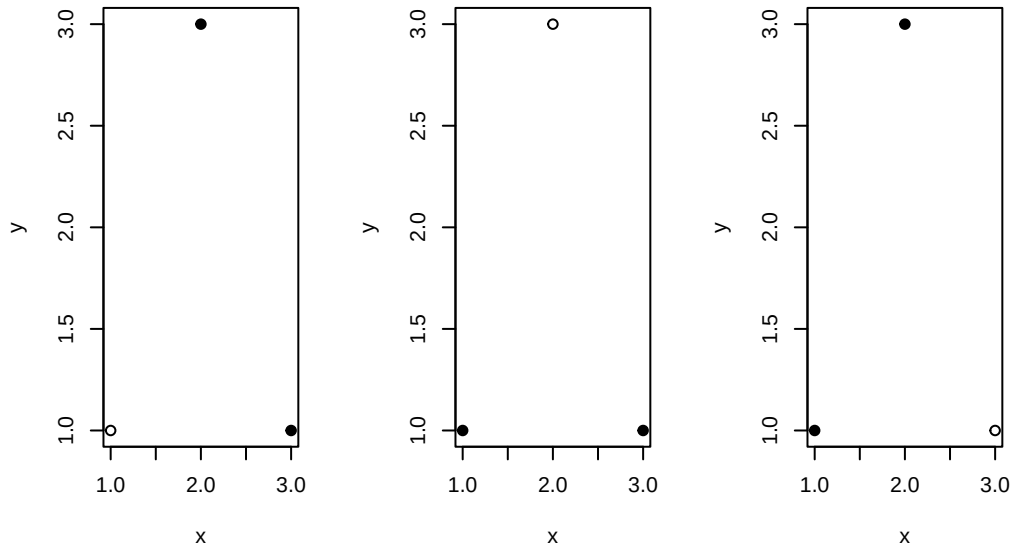
$$\frac{1}{3}[(2 - \frac{2}{3}) + (2.5 - \frac{5}{3}) + (3 - \frac{2}{3})] = \frac{4.5}{3} = 1.5$$

Problem 4

Consider the following data with three observations.

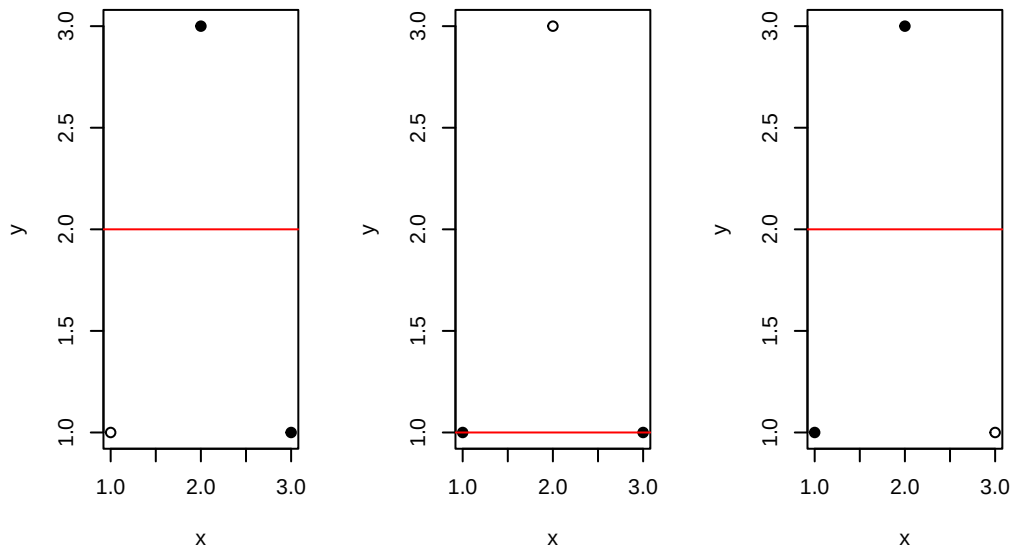


Our goal is to use leave one out cross-validation to choose between a linear regression model and a constant model for this data. Below we have plotted these data three times. In the left panel, the left-most observation has been left out (denoted by an open circle). In the middle panel, the middle observation has been left out. In the right panel the right observation has been left out.



Part A

On the figure above, draw a constant model (i.e., flat line) for each of the three plots based on the non-left-out data.



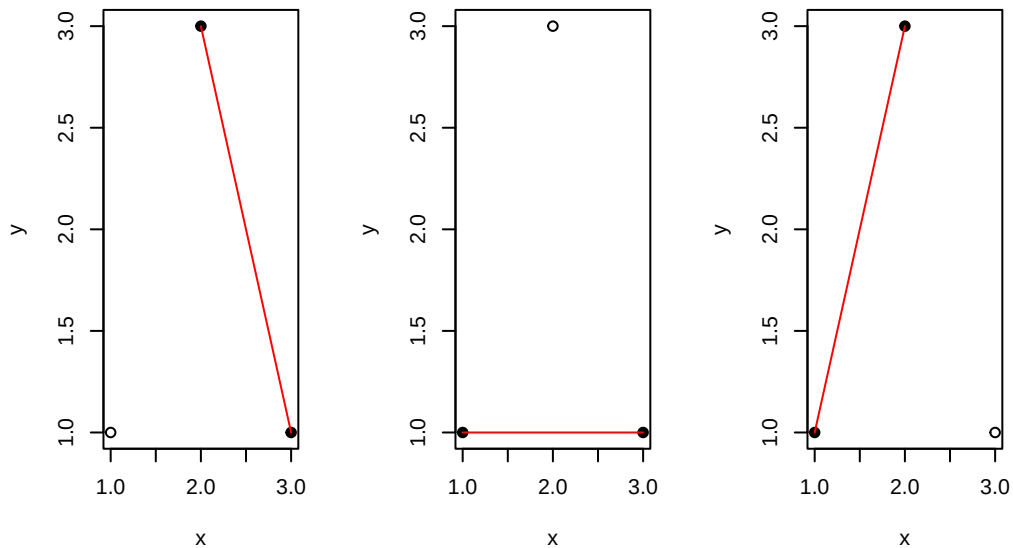
Part B

On the figure above, draw the residuals from the constant models to the left-out observation for each of the three plots.

Left residual = 1; middle residual = 2, right residual = 1

Part C

On the figure above, draw a linear regression line (i.e., line that doesn't have to be flat) for each of the three plots based on the non-left-out data.



Part D

On the figure above, draw the residuals from the linear models to the left-out observation for each of the three plots.

Left residual = 4; middle residual = 2, right residual = 4

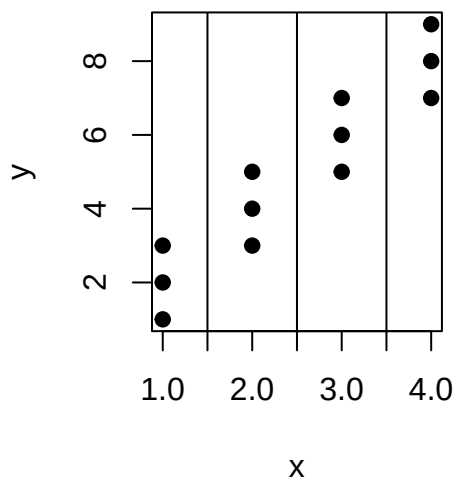
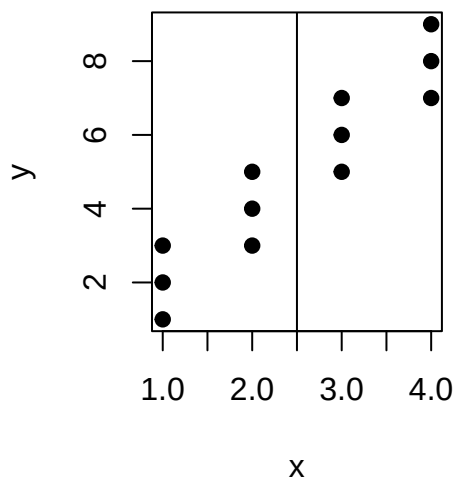
Part E

Based on the residuals you calculated in parts B and D, which model is preferred?

The constant model is preferred to the linear model

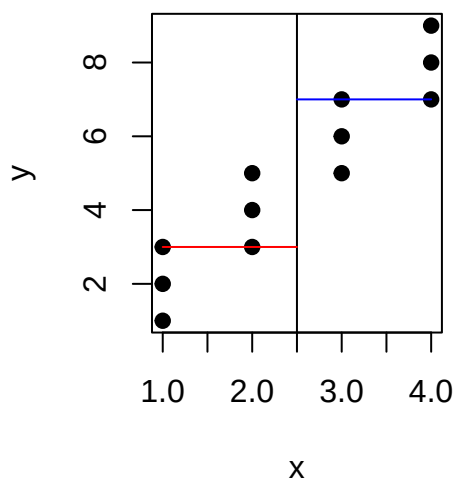
Problem 5

Consider the data below. In the left panel, it has been partitioned into two bins. In the right panel, it has been partitioned into four bins.



Part A

In the left panel, draw piece-wise constant models for the two bins.



Part B

Calculate the estimated standard error for each of the two bins in the left panel. Use the table below to organize your calculations.

Observation i	y_i	bin constant	deviation	deviation ²
1	1	3	-2	4
2	2	3	-1	1
3	3	3	0	0
4	3	3	0	0
5	4	3	1	1
6	5	3	2	4
7	5	7	-2	4
8	6	7	-1	1
9	7	7	0	0
10	7	7	0	0
11	8	7	1	1
12	9	7	2	4

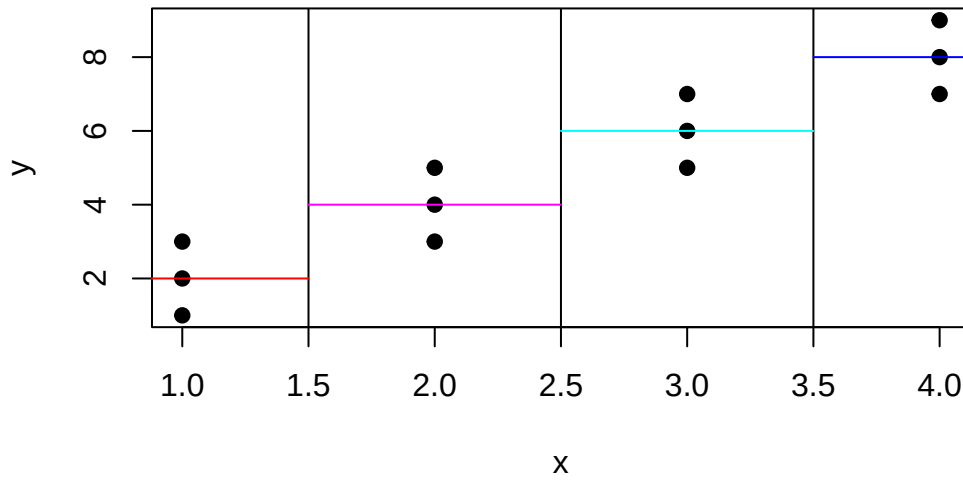
$$\text{sample variance} = \frac{20}{12} = 1.6667$$

$$\text{sample standard deviation} = \sqrt{\frac{20}{12}} = 1.20994$$

$$\text{estimated standard error} = \frac{sd}{\sqrt{12}} = 0.3727$$

Part C

In the right panel, draw piece-wise constant models for the four bins.



Part D

Calculate the estimated standard error for each of the four bins in the right panel. Use the table below to organize your calculations.

Observation i	y_i	bin constant	deviation	deviation ²
1	1	2	-1	1
2	2	2	0	0
3	3	2	1	1
4	3	4	-1	1
5	4	4	0	0
6	5	4	1	1
7	5	6	-1	1
8	6	6	0	0
9	7	6	1	1
10	7	8	-1	1
11	8	8	0	0
12	9	8	1	1

$$\text{sample variance} = \frac{8}{12} = 0.6667$$

$$\text{sample standard deviation} = \sqrt{\frac{8}{12}} = 0.8165$$

estimated standard error = $\frac{sd}{\sqrt{12}} = 0.2357$

Part E

Comment on the difference(s) between the two bin and four bin models.

Problem 5

Part A

Suppose we want β_1 from $y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + e_i$ but we run the regression $lm(y \sim x)$ without z . We have de-meaned our variables x_i and z_i such that $\bar{x} = \sum_{i=1}^n x_i = 0$ and $\bar{z} = \sum_{i=1}^n z_i = 0$. Under what condition(s) will the coefficient on x from $lm(y \sim x)$, which is $\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$, be equal to β_1 ? Derive what your coefficient on x will be in terms of β s and justify your steps.

The coefficient on x from $lm(y \sim x)$ is

$$\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

Plug in y_i defined by the multivariate regression: $y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + e_i$

$$\frac{\sum_{i=1}^n x_i (\beta_0 + \beta_1 x_i + \beta_2 z_i + e_i)}{\sum_{i=1}^n x_i^2}$$

Distribute

$$\frac{\sum_{i=1}^n x_i \beta_0}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i \beta_1 x_i}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i \beta_2 z_i}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i e_i}{\sum_{i=1}^n x_i^2}$$

Let's work through each term.

$$\frac{\sum_{i=1}^n x_i \beta_0}{\sum_{i=1}^n x_i^2} = \frac{\beta_0 \sum_{i=1}^n x_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_0 * 0}{\sum_{i=1}^n x_i^2} = 0$$

as x is demeaned, $\sum x_i = 0$; demeaning is not required for this term to be 0, it just makes it easier

$$\frac{\sum_{i=1}^n x_i \beta_1 x_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_1 \sum_{i=1}^n x_i x_i}{\sum_{i=1}^n x_i^2} = \beta_1$$

$$\frac{\sum_{i=1}^n x_i \beta_2 z_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_2 \sum_{i=1}^n x_i z_i}{\sum_{i=1}^n x_i^2}$$

$$\frac{\sum_{i=1}^n x_i e_i}{\sum_{i=1}^n x_i^2} = \frac{0}{\sum_{i=1}^n x_i^2} = 0$$

we know that $\sum_{i=1}^n x_i e_i = 0$ from the first order conditions on our least squares estimator (we chose the intercept and coefficient(s) that minimize the sum of squared residuals)

Thus we have

$$\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} = \beta_1 + \frac{\beta_2 \sum_{i=1}^n x_i z_i}{\sum_{i=1}^n x_i^2}$$

The coefficient on x from the simple regression of y on x will be equal to β_1 if: β_2 or the coefficient on x from the simple regression from z on x is 0. If either of these terms is non-zero, our simple regression will not give us the same result as our regression that includes both x and z . If x is randomized (like a random treatment assignment), it should not be correlated with z , so our short regression coefficient should give us β_1 .

Part B

We have collected y , x , and a binary variable z . The scatter plots below show our data.

Do you expect the coefficient on x from the simple regression of y on x ($lm(y \sim x)$) to be:

- Positive
- Negative
- Zero

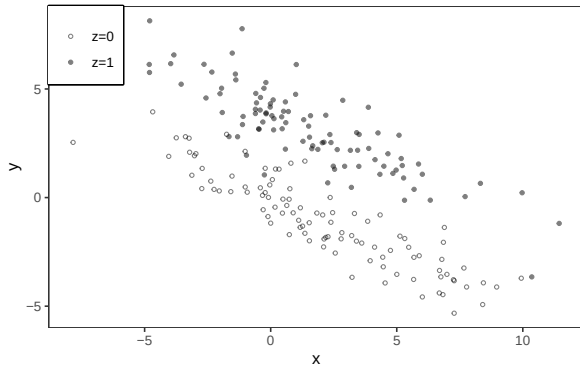


Figure 5: Scatterplot 1

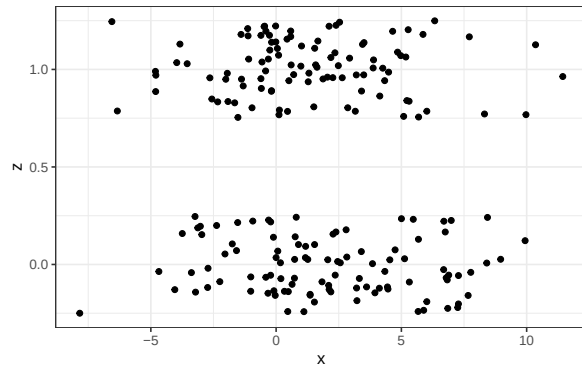


Figure 6: Scatterplot 2 (Jittered)

Part C

We then run the regression $lm(y \sim x + z)$ which gives us the following output:

Call:

```
lm(formula = y ~ x + z, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.9710	-0.6505	-0.0814	0.6073	3.3079

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.00805	0.10772	0.075	0.941
x	-0.50992	0.02014	-25.322	<2e-16 ***
z	3.87919	0.14127	27.459	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9935 on 197 degrees of freedom

Multiple R-squared: 0.8877, Adjusted R-squared: 0.8865

F-statistic: 778.4 on 2 and 197 DF, p-value: < 2.2e-16

And we run the regression $lm(z \sim x)$ which gives us the following output:

Call:

```
lm(formula = z ~ x, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-0.64150 -0.49917 -0.00015  0.48622  0.64447

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.52525     0.03928  13.371  <2e-16 ***
x           -0.01483     0.01008  -1.472   0.143
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4998 on 198 degrees of freedom
Multiple R-squared:  0.01083,    Adjusted R-squared:  0.005832
F-statistic: 2.167 on 1 and 198 DF,  p-value: 0.1426
```

How would you use these outputs to calculate the coefficient on x from the simple regression $lm(y \sim x)$? Explain your steps and define any notation you use. Plug in the estimates from your output (you can leave it as an equation).

From our omitted variable formula (also what we derived in Part A), the coefficient on x from the simple regression of y on x is $\beta_1 + \frac{\beta_2 \sum_{i=1}^n x_i z_i}{\sum_{i=1}^n x_i^2}$ where β_1 and β_2 are from our multiple regression and $\frac{\sum_{i=1}^n x_i z_i}{\sum_{i=1}^n x_i^2}$ is the coefficient on z from the simple regression of z on x .

From our outputs: $\beta_1 = -0.5099$; $\beta_2 = 3.87919$; $coef(z \sim x) = -0.01483$
 $(-0.5099) + (3.87919) * (-0.01483) = -0.5674$

This is close to the β_1 from our multiple regression, as x and z are not strongly correlated, but not exactly the same.

Part D

How do you expect the standard errors of the simple regression of $y \sim x$ to differ from the standard errors of the multiple regression $y \sim x + z$? Why?

Assuming homoskedasticity for simplicity, let s indicate the simple regression of $y \sim x$ and m indicate the multiple regression $y \sim x + z$. β_1 is the coefficient on x in each model.

$$V[\hat{\beta}_1^s] = \frac{\sigma_s^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$V[\hat{\beta}_1^m] = \frac{1}{1-R_{x \sim z}^2} \cdot \frac{\sigma_m^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

The relationship between the variance of β_1 for the multiple regression and the simple regression depends on the relationship between x and z . If x and z are strongly correlated, we are reducing the spread in the x direction by controlling for z , which increases the variability of our estimator. In the formula, $R_{x \sim z}^2$ is close to 1 if x and z are highly correlated. If they are uncorrelated, as they are in this problem, $R_{x \sim z}^2$ is close to 0. Thus the variance inflation factor (left term of equation) is close to 1 and we would expect the standard errors from the multiple regression to be lower than the standard errors of the simple regression as z is predictive of y , but uncorrelated with x .